```
mRunTimes1 = [4.629E-4, 2.9E-6, 1.157E-4, 5.52E-5, 3.816E-4, 8.797E-4, 0.0038416,
0.0113324, 0.1317683, 1.1536602, 47.3603842];
sRunTimes1 = [2.9E-6, 2.2E-6, 1.11E-5, 5.6E-5, 4.527E-4, 4.97E-4, 0.0039575,
0.0239463, 0.0764231, 0.4078273, 2.5969959];

mRunTimes2 = [4.493E-4, 4.6E-6, 4.37E-5, 1.009E-4, 4.206E-4, 0.0015971, 0.0040345,
0.0089946, 0.141613, 2.1541841, 51.4375797];
sRunTimes2 = [4.4E-6, 2.8E-6, 3.17E-5, 8.1E-5, 9.214E-4, 0.0012465, 0.0059157,
0.0307403, 0.0981428, 0.6650865, 2.4936387];

mRunTimes3 = [7.795E-4, 6.2E-6, 2.32E-5, 7.54E-5, 5.154E-4, 0.0012998, 0.0039185,
0.009028, 0.1580892, 2.2890331, 47.6687227];
sRunTimes3 = [7.0E-6, 4.7E-6, 1.68E-5, 1.488E-4, 8.083E-4, 8.33E-4, 0.0152439,
0.0340265, 0.0898652, 0.6804859, 2.6234329];

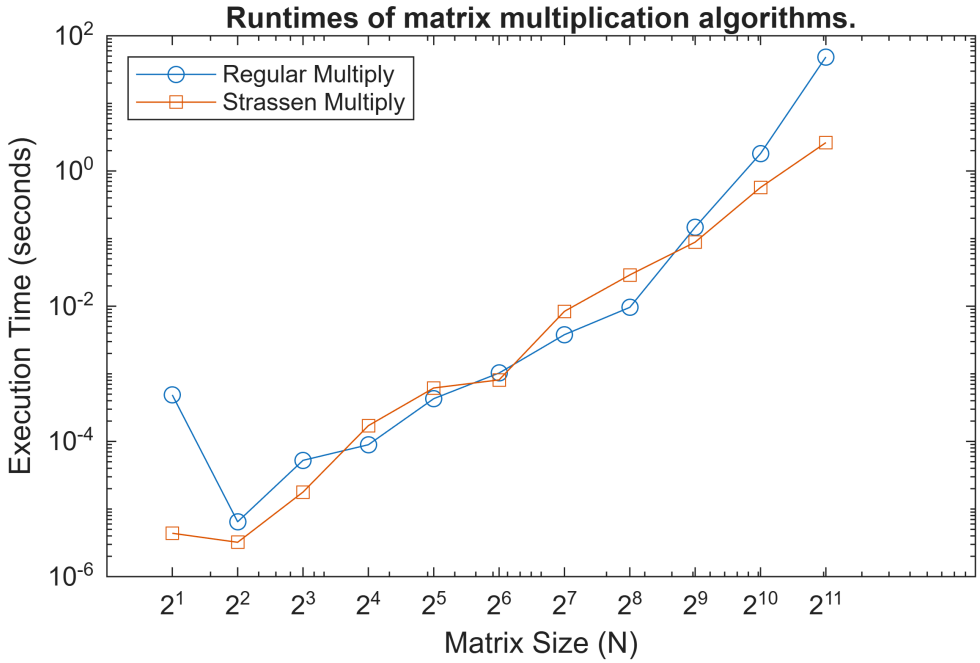
mRunTimes4 = [3.15E-4, 3.3E-6, 5.99E-5, 6.08E-5, 4.267E-4, 7.01E-4, 0.0032606,
0.0107364, 0.1642811, 2.1202796, 50.3043935];
sRunTimes4 = [3.0E-6, 2.5E-6, 1.65E-5, 4.445E-4, 4.409E-4, 9.306E-4, 0.0057745,
0.0344412, 0.0901446, 0.6620373, 2.608414];

mRunTimes5 = [4.306E-4, 1.54E-5, 1.98E-5, 1.543E-4, 3.988E-4, 6.594E-4, 0.003914,
0.0079823, 0.1396482, 1.3382316, 46.0301351];
sRunTimes5 = [4.6E-6, 3.9E-6, 1.3E-5, 1.252E-4, 4.626E-4, 5.572E-4, 0.0110909,
0.0218861, 0.0891013, 0.4368729, 2.8184774];

multiply = [mRunTimes1; mRunTimes2; mRunTimes3; mRunTimes4; mRunTimes5];
strassen = [sRunTimes1; sRunTimes2; sRunTimes3; sRunTimes4; sRunTimes5];

multiply = mean(multiply, 1);
strassen = mean(strassen, 1);

size = [2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048];
figure();
plot(size, multiply, "-o"); hold on;
plot(size, strassen, "-s")
set(gca, "XScale", "log", "YScale", "log");
xticks(size);
xticklabels(arrayfun(@(v) sprintf('2^{%d}', v), [1, 2, 3, 4, 5, 6, 7, 8, 9, 10,
11], 'UniformOutput', false));
xlabel("Matrix Size (N)");
ylabel("Execution Time (seconds)");
legend("Regular Multiply", "Strassen Multiply", "Location", "Northwest")
title("Runtimes of matrix multiplication algorithms.")
```



Write-up:

With the knowledge that the hybrid Strassen multiplication algorithm operates essentially identically to the regular multiplication algorithm for matrix sizes 64 and below, it is no surprise that the run times for the both algorithms are essentially the same for sizes 64 and below. However, for sizes above 64, the performance difference between Strassen multiplication and regular multiplication starts to take form. Between sizes 64 and 512, regular multiplication seems to slightly outperform Strassen multiplication, which isn't surprising considering that Strassen multiplication is more effective for larger matrix sizes. For sizes above 512, Strassen multiplication gains a decisive advantage over regular multiplication, with runtime differences between Strassen and regular multiplications at matrix sizes of 2048 being around 45 seconds apart.